

Engineering Notebook

27518A

Team Number

Rising Phoenixes

Team Name

Highcroft Drive Elementary School

School

08/07/2026

Start Date

End Date



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of

1

Book #

vEx
ROBOTICS

Table of Contents

Topic	Page
Learning, Planning, and Administration	
The Girls Who Built the Bot	3
Goals for this VEXIQ Season	4
Season Schedule & Meeting Calendar	5-6
Build Summary	7
Code Summary and Credit	8
Innovation	9
Engineering Design Process	10
Level Up - VEXIQ Game Review	11
Level Up - Game Strategy	12
Bot 1 - Flop Learn, Design, Build, Iterate	
Days 1-2 Learning parts by building	13-14
Day 3 - Start building Flop	15
Day 4 - Keep building Flop	16-17
Day 5 - Finish building Flop	18-19
Day 6 - Configure Flop	20-22
Day 7 - Field Elements	23
Day 8 - Test Drive Time!	24-25
Day 9 - Brainstorming	26
Bot 2 - Cat Learn, Design, Build, Iterate	
Day 10 - Start building Cat (New Bot)	27-28
Days 11-12 - Build Cat Drivetrain	29-30
Days 13-14 - Drivetrain & Catapult	31-32
Day 15 - Drivetrain gear & Catapult	33-34
Day 16 - Build lifting arm and brain	35
Day 17 - Cat lift motor arm	36-37
Day 18 - Driving Test	38-39
Day 19 - Tweaks to score	40
Day 20 - Cat driving and fixing	41-42
Day 21 - Fix cat arm	43-44
Day 22 - Autonomous Driving Test	45-49
Days 23-24 - Competition Prep	50



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5th grade - 1st year of Vex IQ



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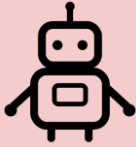
5th grade-1st year of Vex IQ

Goals For This VEX IQ Season

This is our first season working together as a VEX IQ team. We are excited to learn, improve, and have fun while competing in VEX IQ Level Up

Our Big Goal

Since we are all new to VEX IQ, our goal is to learn step by step, help each other, and improve throughout the season.



Robot Building and Design

Learn how to build a strong robot that can pick up, move, and score Bean Bags.



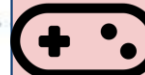
Autonomous Coding

Learn how to code our robot so it can move and score by itself during the Autonomous Skills Challenge.



Engineering Notebook

Write down our ideas, drawings, problems, changes, tests, and what we learned.



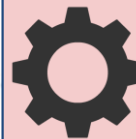
Driving Practice

Practice driving so we can control the robot better, move faster, and score more points.



Game Strategy and Teamwork

Learn the Level Up game rules, plan good scoring strategies, communicate with our teammates, and work well with our alliance partners.



Motors, Sensors, and Coding

Learn how the VEX IQ Brain, motors, sensors, controller, and VEXcode work.

Our Big Goal

Our biggest goal is to keep learning and improving at every practice and competition. We want everyone to get a chance to build, code, drive, share ideas, and solve problems together.

We want to support each other, have fun, and be proud of how much we improve during the season.

Season Schedule & Meeting Calendar

Season: 2026–2027 VEX IQ | Team meets 3x per week
Fridays 5:30-7 | Saturdays and Sundays 2-4

Date	Project Day	Page Number	Day of Week	Activity / Focus
August 8, 2026	Day 1	13-14	Saturday	Learning parts by building
August 11, 2026	Day 2	13-14	Tuesday	Learning parts by building
August 14, 2026	Day 3	15	Friday	Start building Flop
August 15, 2026	Day 4	16-17	Saturday	Keep building Flop
August 16, 2026	Day 5	18-19	Sunday	Finish building Flop
August 21, 2026	Day 6	20-22	Friday	Configure Flop
August 22, 2026	Day 7	23	Saturday	Field Elements
August 23, 2026	Day 8	24-25	Sunday	Test Drive Time!
August 28, 2026	Day 9	26	Friday	Brainstorming
August 29, 2026	Day 10	27-28	Saturday	Start building Cat (New Bot)
August 30, 2026	Day 11	29-30	Sunday	Build Cat Drivetrain
September 4, 2026	Day 12	29-30	Friday	Build Cat Drivetrain
September 11, 2026	Day 13	31-32	Friday	Drivetrain & Catapult
September 12, 2026	Day 14	31-32	Saturday	Drivetrain & Catapult

Team: Rising Phoenixes | 27518A | Highcroft Drive Elementary School | 2026 - 2027

Season Schedule & Meeting Calendar

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Date	Project Day	Page Number	Day of Week	Activity / Focus
September 13, 2026	Day 15	33-34	Sunday	Drivetrain gear & Catapult
September 14, 2026	Day 16	35	Saturday	Build lifting arm and brain
September 20, 2026	Day 17	36-37	Sunday	Cat lift motor arm
September 21, 2026	Day 18	38-39	Monday	Driving Test
September 22, 2026	Day 19	40	Tuesday	Tweaks to score
September 23, 2026	Day 20	41-42	Wednesday	Cat driving and fixing
September 25, 2026	Day 21	43-44	Friday	Fix cat arm
September 26, 2026	Day 22	46-49	Saturday	Autonomous Driving Test
September 27, 2026	Day 23	50	Sunday	Competition Prep
September 28, 2026	Day 24	50	Monday	Competition Prep
September 29, 2026	Day 25	TBA	Tuesday	Competition Prep
September 30, 2026	Day 26	TBA	Wednesday	Competition Prep
October 1, 2026	Day 27	TBA	Thursday	Competition Prep
October 2, 2026	Day 28	TBA	Friday	Competition Prep
October 3, 2026		TBA	Saturday	COMPETITION DAY!

Team: Rising Phoenixes | 27518A | Highcroft Elementary School | 2026 - 2027

Build Summary

What We Learned

This is our first VEX IQ season. We started by building Flop to learn about robot parts, gears, motors and coding. After testing Flop, we found problems with its balance, arm support and clearance. We learned that fixing one problem can sometimes create another problem. We compared two robot designs and decided to build our new robot, Cat.

Skills We Built

We learned how to build drivetrains, use different gear ratios, connect motors and configure our controller. We also learned how to solve problems with gears, chains and the catapult. We practiced block coding and created an autonomous program that completed multiple cycles during testing.

What Our Robot Can Do

Our robot, Cat, has a drivetrain, an intake system and a rubber-band-powered catapult. We tested different rubber band combinations until we found one that could shoot bean bags to Level 3. We are still working on making the bean bags stay on Level 3 instead of bouncing off. Our autonomous program uses a bumper sensor to stop the catapult and a loop to repeat its movements.

Engineering Highlights

Our catapult had several problems during testing. The rubber bands slipped off because the shaft was too short, so we replaced it with a longer shaft. We tested different rubber band combinations to help us reach Level 3. Sometimes the catapult also moved too far backward and got stuck. We added a stopper to prevent this from happening. We also designed two antennas and added supports to slightly tilt our robot to help improve scoring.

Code Summary and Credit

Our Autonomous Program

Our program sets the driving, turning, arm, intake and catapult motor speeds before starting.

We use a variable called autonLoop. The robot follows a slightly different driving path during the first cycle compared with later cycles.

The robot starts the catapult and waits until the bumper sensor is pressed. Once the bumper is pressed, the catapult motor stops.

The program moves the intake and arm, drives through its programmed route, and prepares the catapult for another cycle.

After each cycle, the program increases autonLoop and repeats the process.

Credits

VexIQ for building our First robot flop

2024–2025 Rapid Relay robot: Inspiration for our slingshot-style launcher design.

2023–2024 Clawbot: Inspiration for our intake/claw design.

Mentor assistance: We received training from Aarush (Ruhi's brother) while learning robot parts and building our first robot. He also helped us mentor during the season.

Youtube videos: We watched many videos on youtube.

Innovation

Problem: We needed a way to automatically stop our catapult at the correct position so it would be ready for the next cycle.

Our Innovation: We used a bumper sensor with block coding. When the catapult presses the bumper, the sensor sends a signal to the brain, and our program stops the catapult motor.

How It Works:

The program starts the catapult motor.

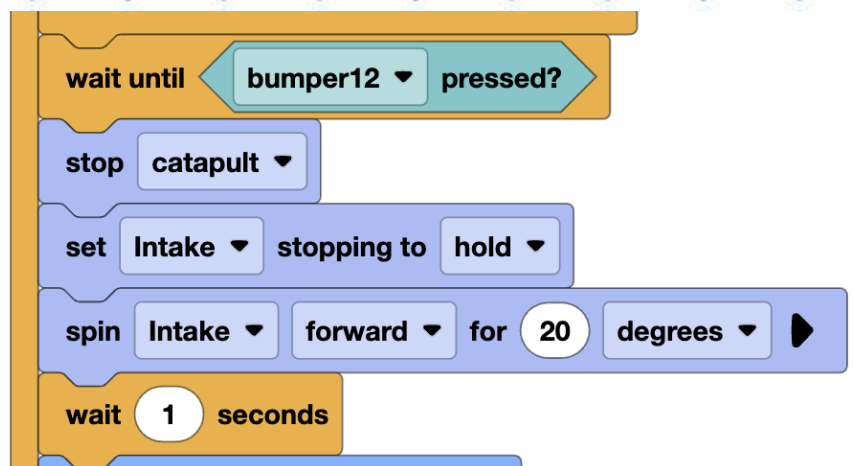
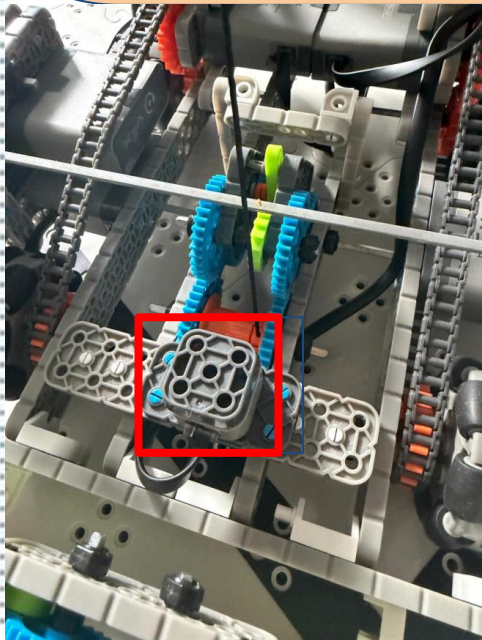
The program waits until the bumper sensor is pressed.

When the bumper is pressed, the catapult motor stops.

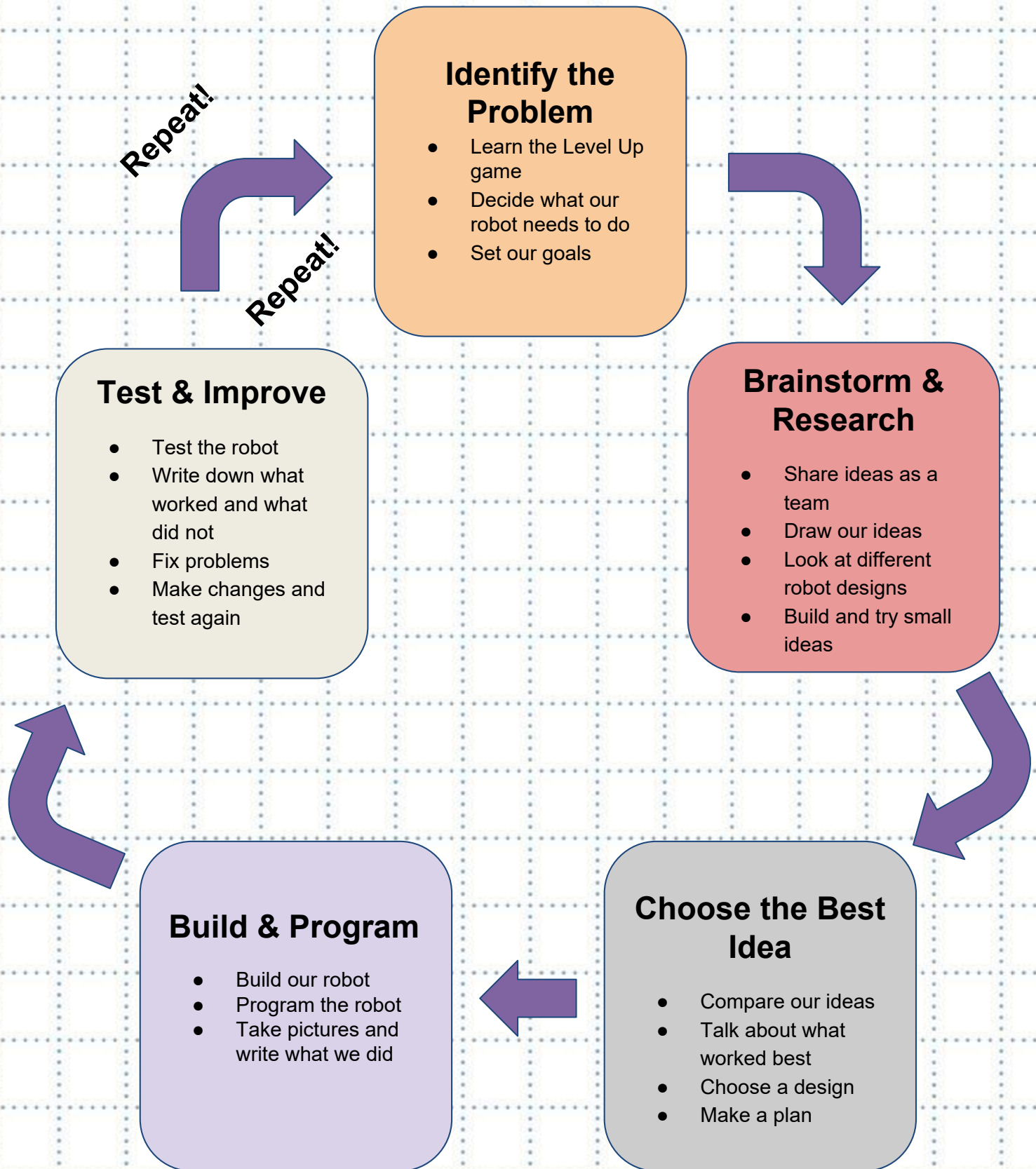
The robot continues with the next steps of its autonomous program.

Testing Result: Our autonomous program successfully completed multiple cycles during testing.

What We Learned: We learned how to use a sensor and coding together to control a moving part of our robot instead of relying only on motor timing.



Engineering Design Process



Level Up - VEX IQ Game Review

Level Up is a VEX IQ game where robots pick up bean bags and score them in different goals



Bean Bags: Red, blue, and yellow game pieces. A robot can carry only **one** bean bag at a time.

Load zones

Area where bean bags can be loaded during the match. Each robot starts with one

Scoring

- Floor Goal = 1 point
- L4 = 16 points (yellow only!)

Yellow bean bags can be scored in any color goal and L4

Blue bean bags can be scored in blue goals

Red bean bags can be scored in red goals

6 feet

8 feet

Match Types

Teamwork Challenge: Two robots work together for **60 seconds**.

Driving Skills: One robot is driver-controlled for **60 seconds**.

Autonomous Coding Skills: One robot runs using code for **60 seconds**.

Our goal is to score as many points as possible by choosing the best Bean Bags, Goals, and driving paths while working well with our teammate.

The best Driving Skills and Autonomous Coding Skills scores are added together for the Robot Skills score.

Scoring

- L1 = 3 points
- L2 = 6 points
- L3 = 12 points

Robot Size Rules

Starting size: Maximum 11" x 20" x 15".

Horizontal expansion: Maximum 11" x 24" during the match.

Vertical expansion: No height limit after the match starts.

Level Up - Game Strategy

First strategy thoughts

Higher Goals give more points, so we need to decide if we should score quickly in lower goals or spend more time reaching L2, L3, or L4. Our robot can carry only **one Bean Bag at a time**, so choosing short and fast paths is important. Yellow Bean Bags are useful because they can score in either color Goal and are the only Bean Bags that can score on L4. ([VEX Robotics](#))

For the first competition

We decided to use the Cat robot. We drive the Cat to line up straight with the hurdle field element (lines up with our bumpers) and catapult the bean bags from there. It is a straight line drive so it is fast and easy. We hope to hit level 3 every time, but will be okay if it hits level 2 every time.

This is where we will talk about what worked and what didn't after the first competition.

For the second competition

We decided to follow strategy XXXXX.

This is where we will talk about what worked and what didn't after the third competition.

For the third competition

We decided to follow strategy XXXXX.

This is where we will talk about what worked and what didn't after the second competition.

For the fourth competition

We decided to follow strategy XXXXX.

This is where we will talk about what worked and what didn't after the third competition.

Days 1-2 - Learning parts by building

Goal: to learn about the parts

Action Plan

1. Getting to know about VEX IQ
2. Learning about each part

Accomplishments

- making the drive train
- getting training from mentor
- learning something new
- coded brain to make gears turn

Challenges

- Finding and remembering the names of robot parts
- Finding out how to perfectly line up the pieces
- Finding out how to fix the mistakes that we made

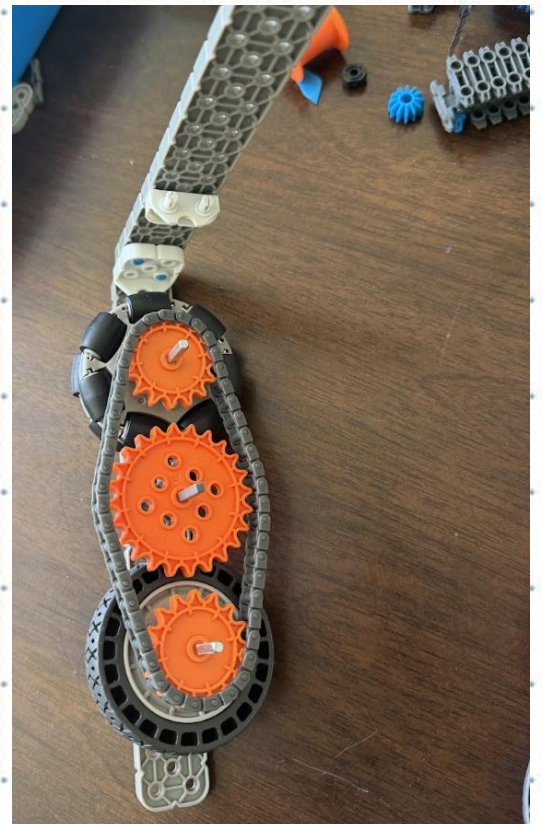
Lessons learned

- gear ratio
- learning about brain and how to configure it
- learning about motor and wheel types

Next steps

Build our favorite robot - Flop

Days 1-2 - Learning parts by building



Project VexIQ Introduction

Day 3 - Start building Flop

Goal: to get to step 50

Action Plan

1. Work without distractions
2. Focus on building the bot
3. Stay on one topic not talking with your friends about something non VEXIQ-related
4. 2 Members part fetchers and 2 members build

Accomplishments

- We Completed Up to Step 23

Challenges

- Attaching the parts to the other parts because they were tight
- Finding where the pieces lined up
- Finding the right parts from so many parts

Lessons learned

- We learned more about each part
- We learned more about what each part was called and how we use it
- How standoff works
- Difference between capped pins and non-capped pins
- How to build it like a lego

Next steps

Finish building the arm

Day 4 - Keep building Flop

Goal: Complete building bot arm

Action Plan

1. 2 teams will focus on different parts of the robot
2. We will build until arm is complete

Accomplishments

- We completed the arm

Challenges

- Dividing the work
- Coordinating with the other team when their part was completed
- Putting parts together to build the arm
- We made the intake wrong and had to fix it

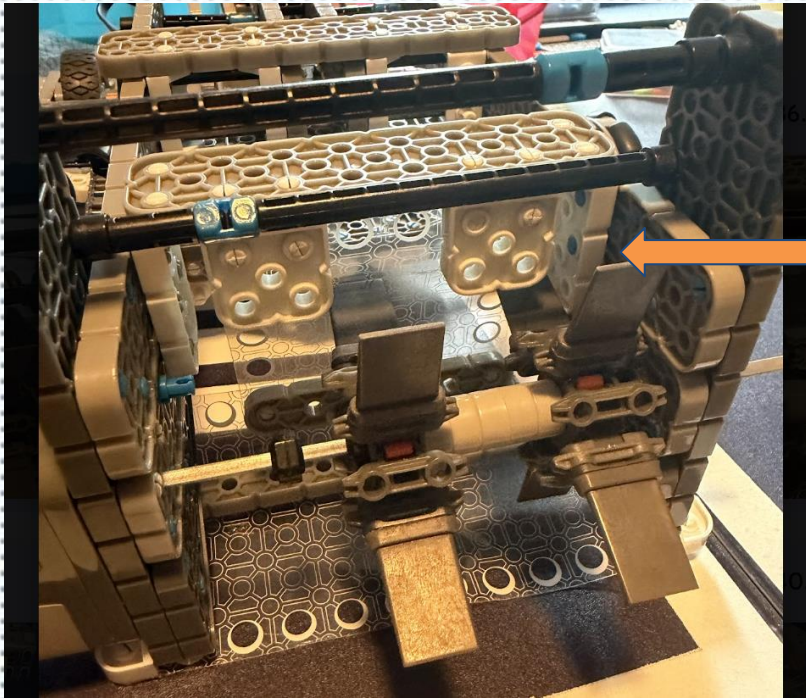
Lessons learned

- We learned that we made better progress by dividing into two teams
- We learned to work faster and better
- We learned to carefully follow the steps or we have to take it apart to fix it
- We learned about gear ratio and how to identify different gear sizes

Next steps

Complete the bot

Day 4 - Keep building Flop



Problem: The white spacers were installed on the front side of the intake shaft, causing the intake rollers spinning freely. Since we had already attached plastic sheet on the back it was very challenging to undo, so had to take apart few things and put it back.

Solution: We moved the spacers to the correct side and checked that each intake roller could rotate freely by hand.

Finished building the flop arm!



Day 5 - Finish building Flop

Goal: Finish building Flop

Action Plan

1. Complete focus on building
2. Complete building Flop

Accomplishments

- We made the drive train
- We make flop arm rest so it does not touch the ground
- We attached everything together to complete all the steps

Challenges

- We installed bumper sensor, but we don't know how to use it.
- Wiring was difficult as arm would move when we flipped, and some places it was too narrow to put our fingers to insert cable

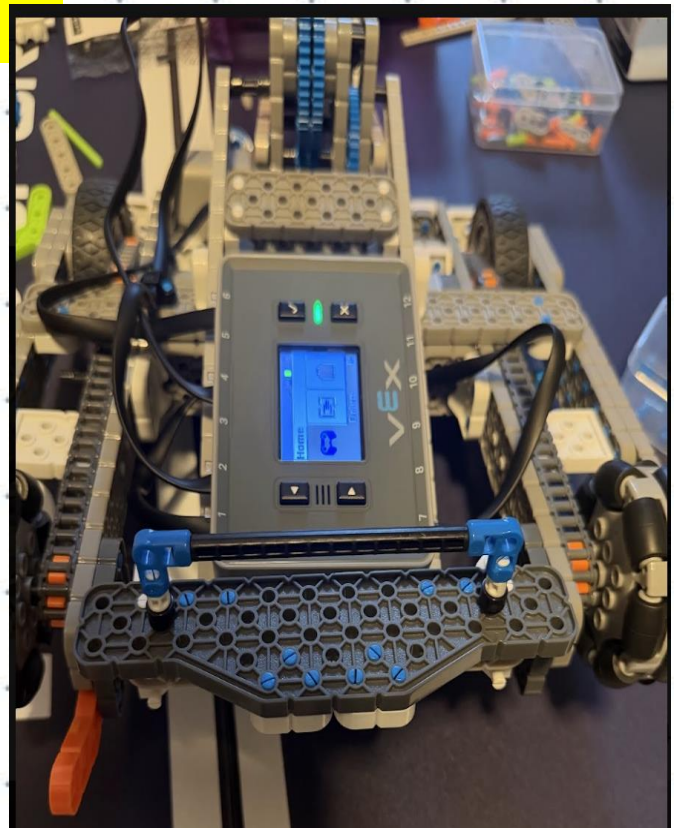
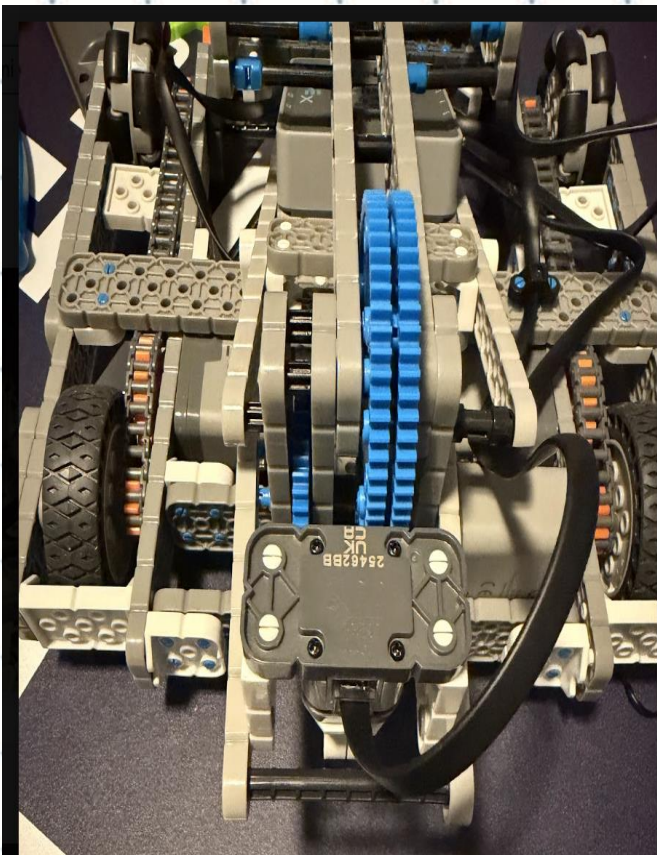
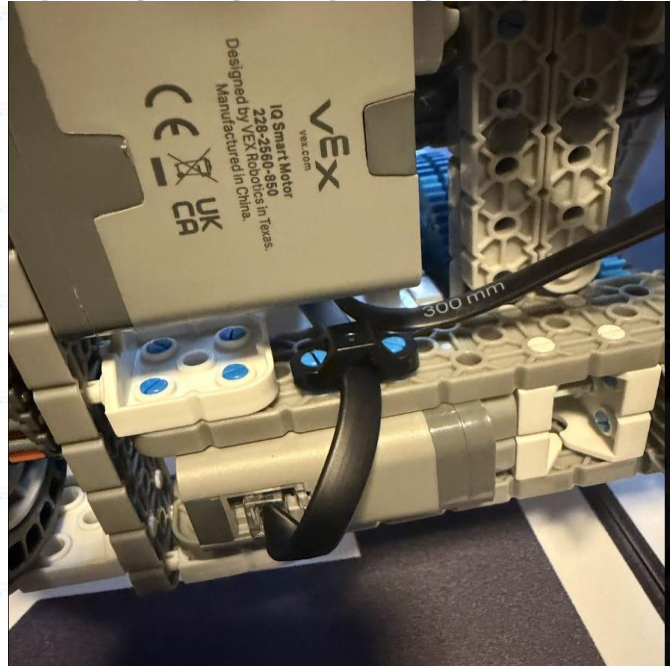
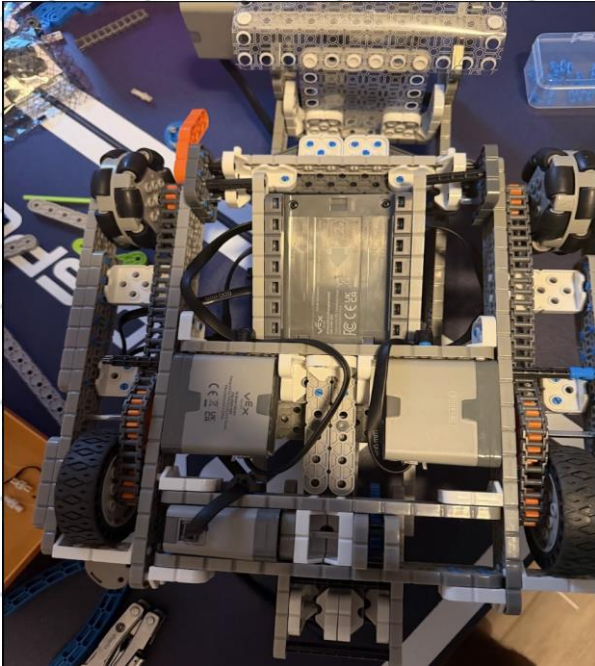
Lessons learned

- How we could build better and faster
- How our engineering notebook is helping us to stay on the task and set goals
- Cable length and how to ping wire so it does not touch the ground

Next steps

1. Program flop and test drive
2. Learn the buttons on the controller
3. Learn the actions flop can do

Day 5 - Flop Completed



Day 6 - Configure Flop

Goal: Configure Flop so we can start driving

Action Plan

1. Configure all ports
2. Start driving practice

Accomplishments

- Configured different ports of brain
- Fixed 3 issues
- Fixed weight imbalance
- Fixed chain skipping
- Fixed armrest snapping

Challenges

- Driving was difficult
- Chain was skipping
- When we flipped arm it would snap the arm rest
- When arm moved to other side wheels will go up

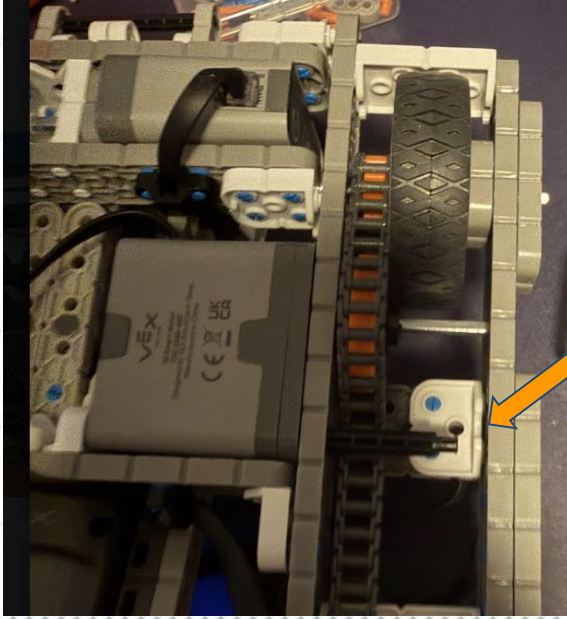
Lessons learned

- Flop basic instructions aren't enough to start driving
- Chain needs the right amount of tightness so it does not skip.
- Parts need to be strong and supported
- Fixes are not easy
- How to configure drive train, arm, and intake roller

Next steps

1. Build the field elements
2. Driving practice to see how many points we could score

Day 6 - Challenges and Solutions



Chain Skipping

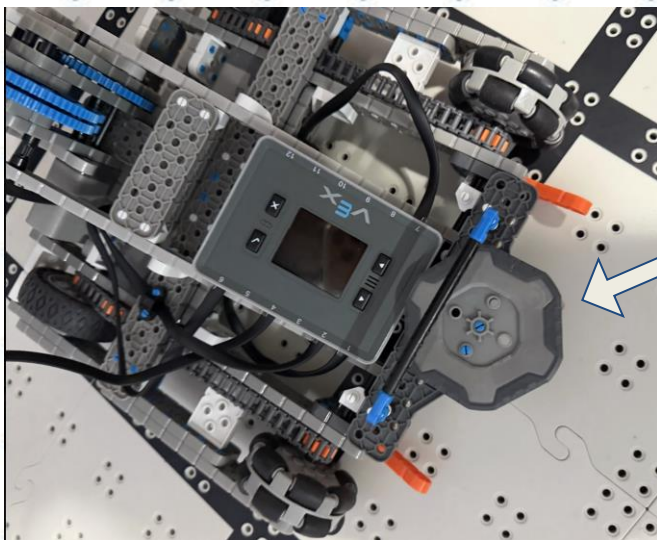
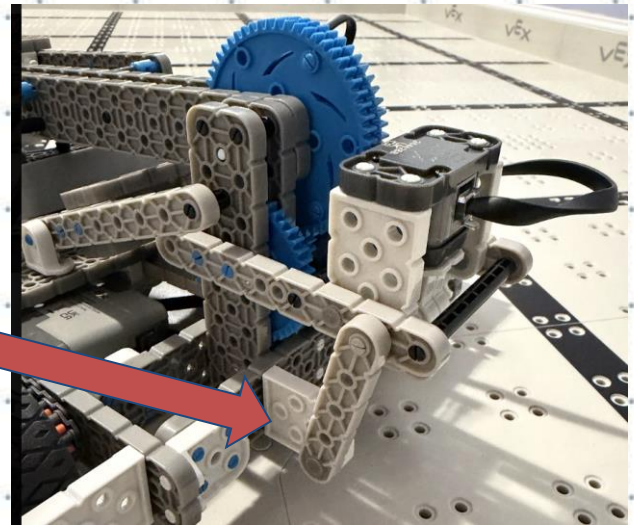
Problem: While we were driving the robot, the chain kept skipping. We found that the standoff was not fully connected to the other side, which made the chain loose. After fixing that, the chain was still skipping.

Solution: We tried removing one chain link to make the chain tighter. After removing the link, the chain stopped skipping and worked better.

Arm Support

Problem: The arm rest was only connected to the side bar as per flop design instructions. When the lift came down and rested on it, the piece would snap off.

Solution: We added another white piece and a 1x4 beam to make it stronger and give it more support.

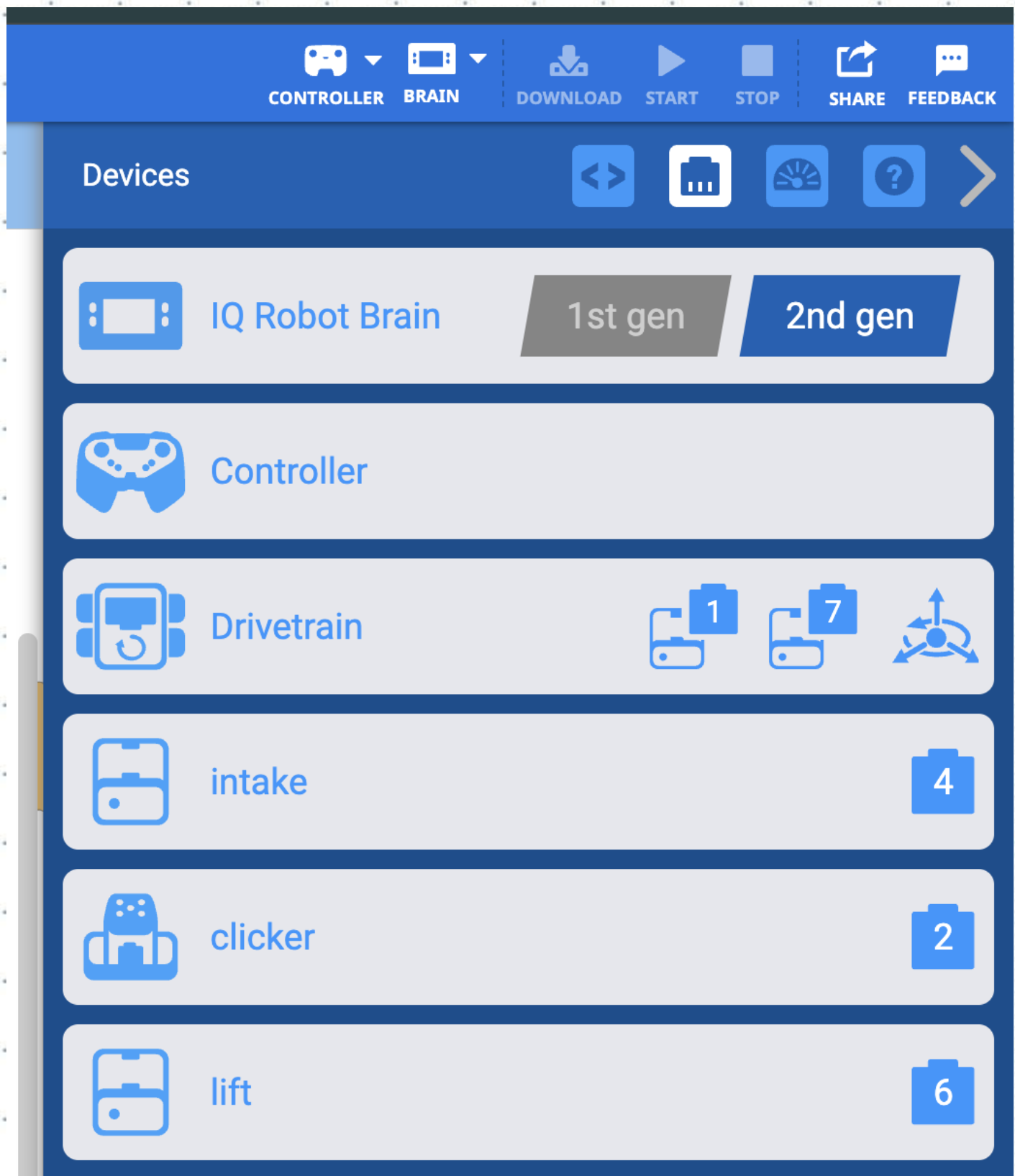


Weight Balance

Problem: When the lift moved, the robot became unbalanced and the opposite side of the robot started lifting off the ground.

Solution: We added a counterweight to the opposite side of the robot to balance the weight.

Day 6 - Flop Motors, Arm and Intake



Day 7 - Field Elements

Goal: Test drive flop

Action Plan

1. Build field elements so we can test drive Flop

Accomplishments

- Built one side of the field elements so we could test drive

Challenges

- Too many steps to follow

Lessons learned

- Teamwork

Next steps

1. Finish field elements
2. Test drive flop

Day 8 - Test Drive Time!

Goal: Test drive flop

Action Plan

1. Test drive Flop
2. Fix any issues

Accomplishments

- We successfully completed our first test drive!

Challenges

- When we pressed the up button, the robot moved backward instead of forward.
- Flop couldn't reach level 3.
- Our robot got stuck on the yellow hurdle on the floor.
- Our robot got stuck when approaching target because the arm was too low.

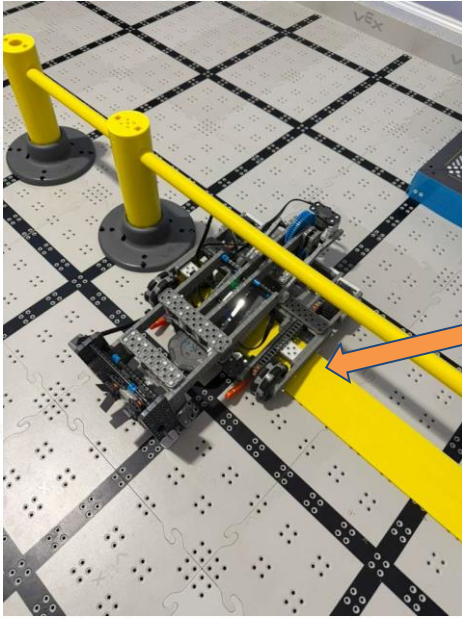
Lessons learned

- We learned how to configure and control our robot.
- We learned that testing helps us find problems we did not notice while building.
- We realized that the robot's arm position and clearance are important.

Next steps

1. Fix the controller settings
2. Check and improve the robot's clearance for the yellow hurdle
3. Find a way to lift the arm when approaching the target
4. Test the robot again

Day 8 - Challenges and Solutions



Clearance

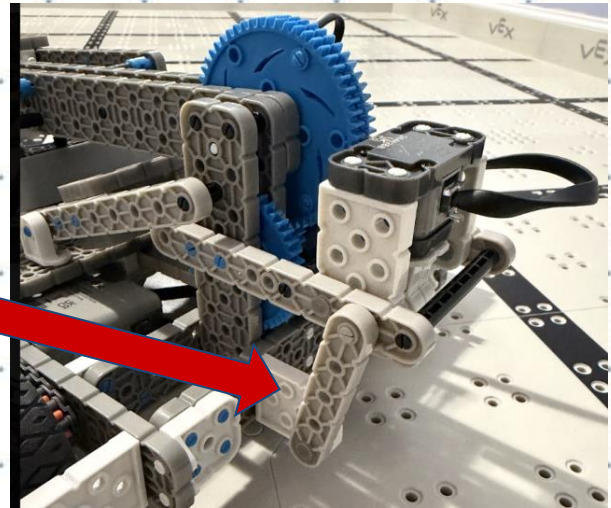
Problem: Our robot got stuck on the yellow hurdle.

Solution: We found that lifting the arm slightly before reversing helped the robot move backward without getting stuck.

Arm Support

Problem: We had added extra support to make the arm rest stronger. During testing, we found that the support stopped our lift from reaching Level 3. It also reduced the clearance under the yellow bar.

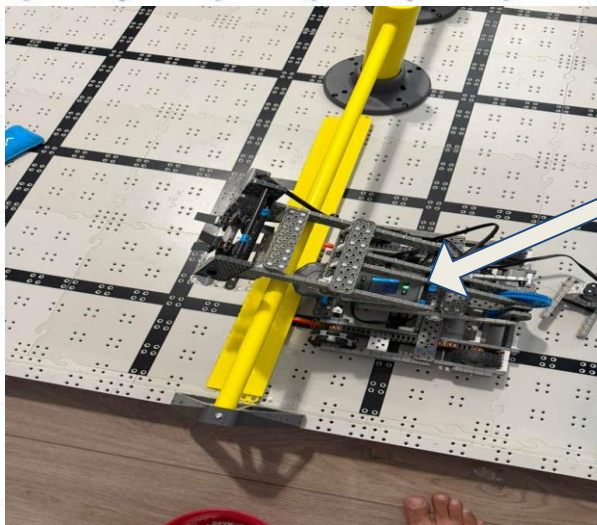
Solution: We removed the extra support because we learned that the Level 1 base could support the arm. However, we need to remember to move the lift to the other side before driving backward so it does not get stuck.



Arm Getting Stuck While Reversing

Problem: After removing the extra support, we found that if we forgot to move the arm before reversing, the arm rest could snap off. Even if we moved the robot backward a little and then tried to move the arm, it would get stuck against the yellow bar.

Solution: We practiced moving the arm to the other side before reversing the robot. We learned that the order of our movements is very important to prevent the arm from getting stuck or breaking.



Day 9 - Brainstorming

Goal: Decide to improve Flop or build a new bot

Action Plan

1. Come up with pros and cons of flop

Accomplishments

- We made the decision together to build the cat

Challenges

- When we pressed the up button, the robot moved backward instead of forward.
- Flop couldn't reach level 3.
- Our robot got stuck on the yellow hurdle on the floor.
- Our robot got stuck when approaching target because the arm was too low.

Lessons learned

- How to brainstorm with a team

Next steps

1. Start building new bot

Day 10 - Start Building Cat (New Bot)

Goal: Building Catapult (Cat)

Action Plan

1. Team 1: Build the mechanism for the launcher
2. Team 2: Build Claw

Accomplishments

- We took inspiration from 2024–2025 Rapid Relay robot inspired a slingshot-style launcher design to build over launcher mechanism
- We took inspiration from 2023-2024 Clawbot to design our intake system

Challenges

- We aligned the claw in the middle, but after sometime we noticed claw was sideways
- Catapult mechanism gears did not align

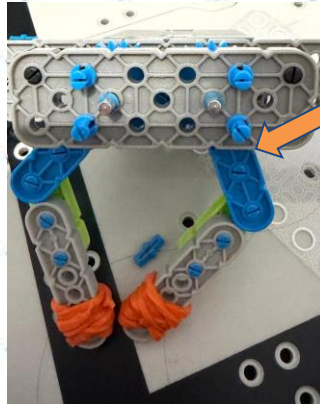
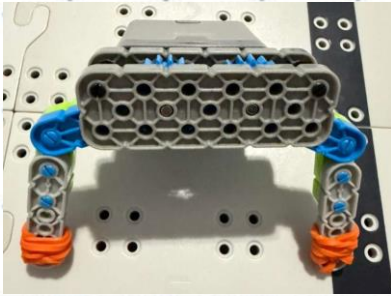
Lessons learned

- All the gears must be properly aligned, or they can jam and create friction
- We learned how gears can transfer motion and how rubber bands can store energy.
- Learned about chunking strategy

Next steps

1. Start Cat build for competition
2. Build drivetrain

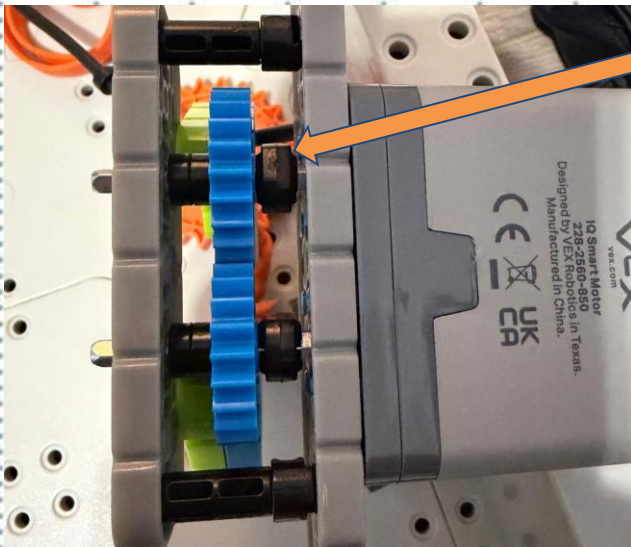
Day 10 - Challenges and Solution



Claw Flaw

Problem: We aligned the claw in the middle, but after sometime we noticed claw was sideways. We that there was too much space between the two plates, causing the gears to slide up and down on the shafts and move out of position.

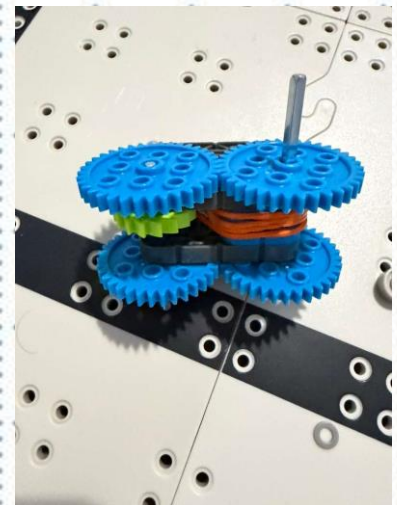
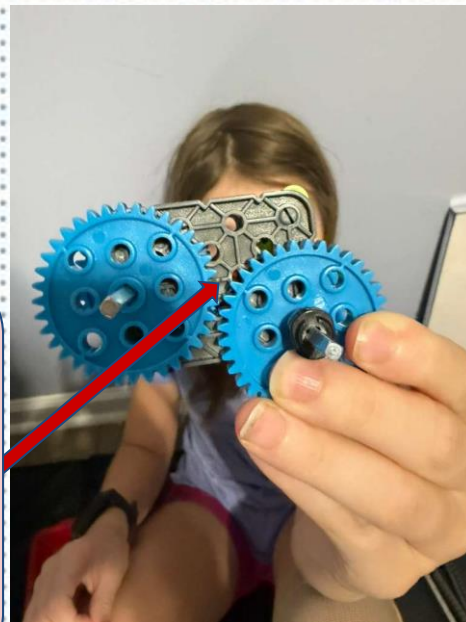
Solution: We adjusted the plates to remove the extra space and added rubber shaft collars to keep the gears in place.



Catapult Mechanism

Problem: There was small space between the two gears, so one gear could not turn the other.

Solution: We moved one gear by one hole so the gears could connect and turn together.



Days 11 & 12 - Build Cat Drivetrain

Goal: Build drivetrain

Action Plan

1. Team 1: Build one side of drive train
2. Team 2: Make other side of the drivetrain

Accomplishments

- Completed building both sides of the drivetrain
- Created enough space for the motor placement for the catapult mechanism

Challenges

- Where to put the motors for catapult mechanism
- Where to put the motors for the drivetrain

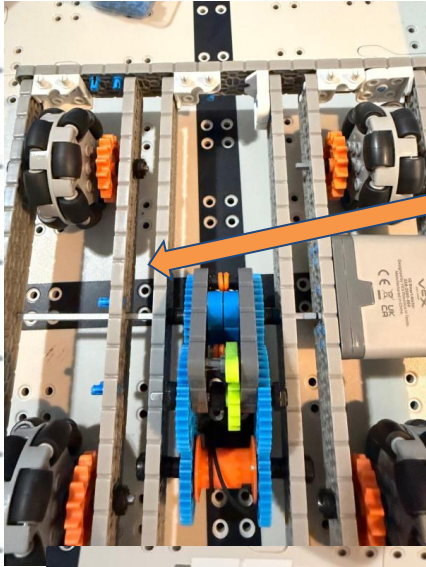
Lessons learned

- We learned how to leave enough room for the motors
- Think ahead of next things so we don't have to redo

Next steps

1. Finish building drivetrain

Days 11 & 12 - Challenges and Solution

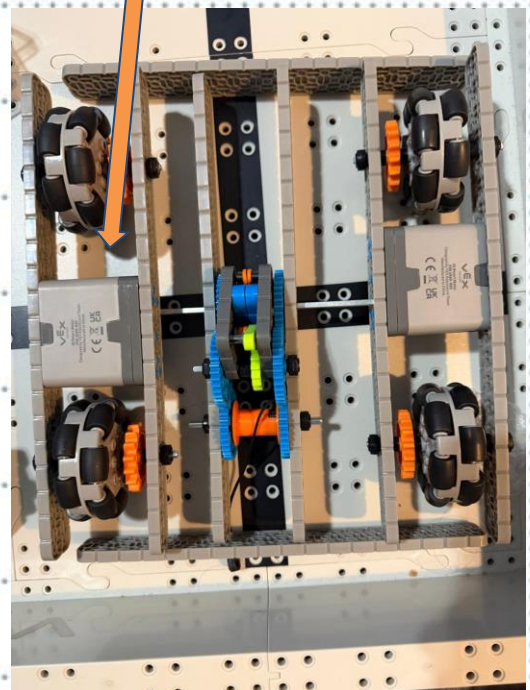
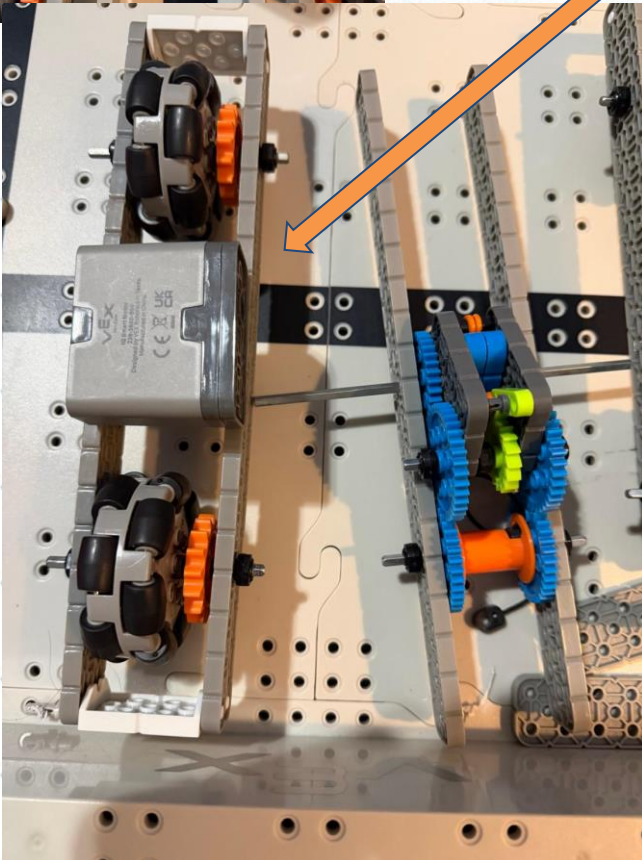


Motor Placement

Problem 1: There was not enough space between the wheels to install the catapault motor. Placing the motor between the catapault mechanism and drivetrain the robot to exceed our 11-inch size requirement.

Problem 2: There was not enough space between the wheels to put the motor.

Solution: We adjusted the spacing between the wheels to create enough room for the catapault motor while keeping the robot within the size limit.



Days 13 & 14 - Drivetrain & Catapult

Goal: Assemble drivetrain and work on catapult

Action Plan

1. Team 1: Build one side of drive train
2. Team 2: Make other side of the drivetrain

Accomplishments

- Completed building both sides of the drivetrain
- Created enough space for the motor placement for the catapult mechanism

Challenges

- Where to put the motors for catapult mechanism
- Where to put the motors for the drivetrain

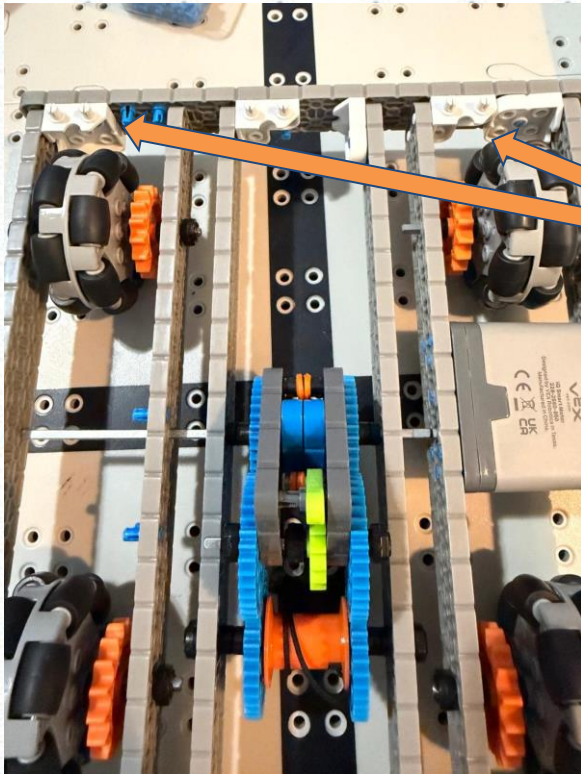
Lessons learned

- We learned how to leave enough room for the motors
- Think ahead of next things so we don't have to redo

Next steps

1. Finish building drivetrain

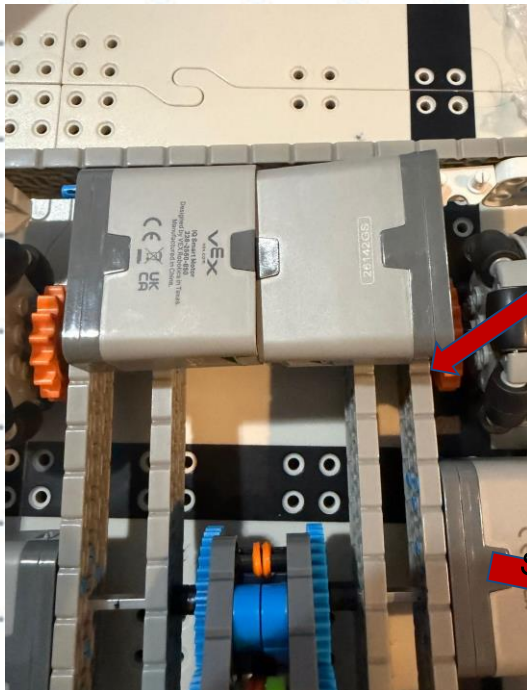
Days 13 & 14 - Challenges and Solution



Drivetrain Assembly

Problem: While assembling the drivetrain, we had trouble inserting the connector pins because some of the holes did not line up.

Solution: We tried different ways to connect the pieces until all the holes lined up and we could insert the pins correctly.

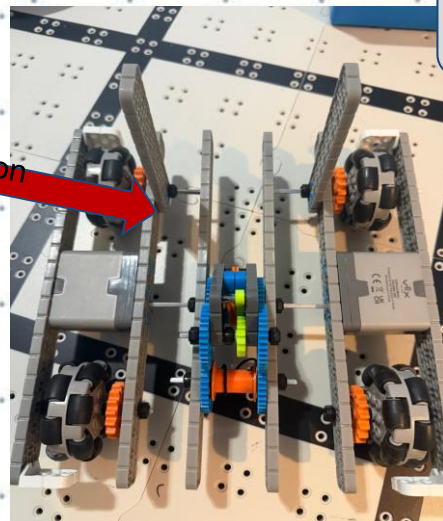


Drivetrain Motor

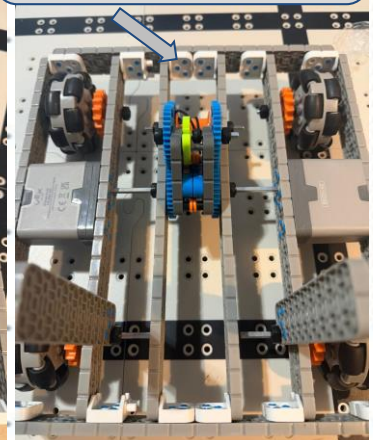
Problem: When we tried to attach the motors to our drivetrain, we realized there was not enough space for them to fit.

Solution: We took apart part of the drivetrain and rebuilt it with extra spacing so the motors could fit properly.

Solution



Redo of Connector Pins



Day 15 - Drivetrain gear & Catapult

Goal: Complete drivetrain and Catapult

Action Plan

1. Put the drivetrain gear
2. Attach catapult to the bot

Accomplishments

- We put the motor above the drivetrain.
- We used a chain and sprockets to connect the motor.
- We fixed the sprocket alignment using black spacers.
- We found the right bar and string lengths for the catapult.

Challenges

- Finding a way to connect the motor to the drivetrain.
- The two sprockets did not line up.
- Finding the right bar and string lengths.
- The string was too loose.

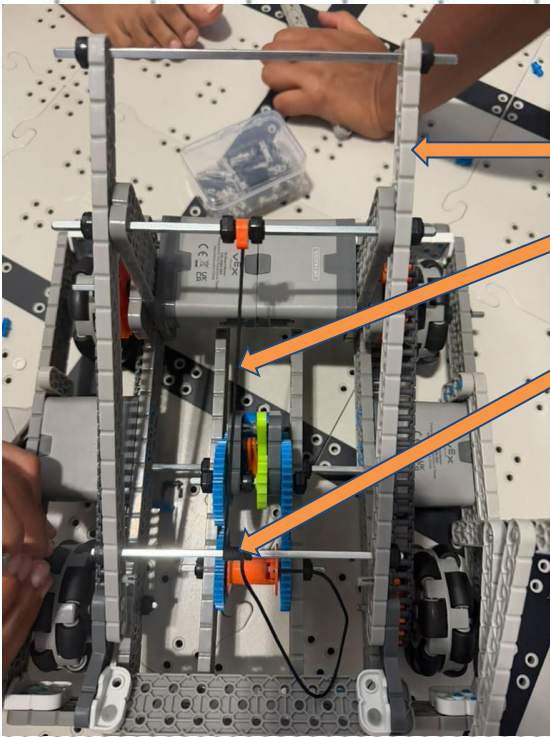
Lessons learned

- We learned that sprockets need to line up properly.
- We learned that string length and tension matter.
- We learned that trying different designs helps us improve our robot

Next steps

1. Start competition build!

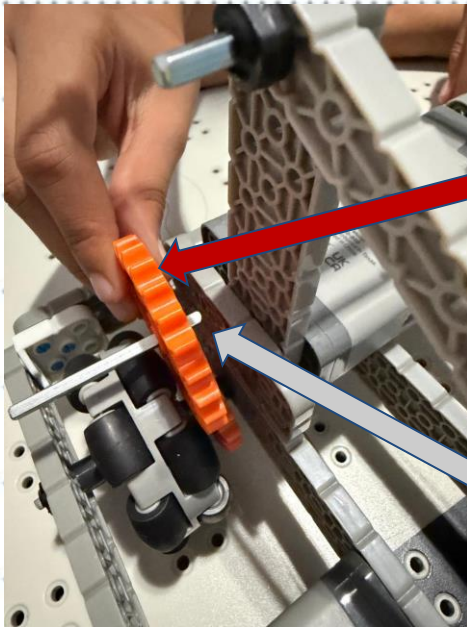
Day 15 - Challenges and Solutions



Catapult Bar Length and String Length

Problem: We needed to find the correct bar length for the catapult, so we tested several different lengths. Our kit did not include the string, so we had to order it separately. When it arrived, we had to choose the correct length, but there was still extra slack.

Solution: We tested different bar lengths to find the right fit and selected the most suitable string length. We then wrapped the extra string around one of the shafts to remove the slack and create proper tension.



Motor Placement

Problem: We wanted to mount the motor above the drivetrain while still transferring power to the wheels.

Solution: We attached a sprocket to the motor and connected it to the drivetrain sprocket using a chain.

Redo of Connector Pins

Problem: The motor sprocket and drivetrain sprocket were not properly aligned to put the chain we needed to be aligned

Solution: We added black spacers to align both sprockets so the chain could move smoothly.

Day 16 - Build lifting arm and brain

Goal: Complete building arm for intake and brain placement

Action Plan

1. Team1: Work on building arm for intake
2. Team 2: Work on placement of robot brain

Accomplishments

- Finished putting brain
- Created lift mechanism

Challenges

- Finding the right place to install the brain on our robot.
- It was difficult to reach some tight spaces to attach the brain's support bar.

Lessons learned

- We learned how to build a lift mechanism.
- We learned that putting parts in tight spaces can be tricky.

Next steps

1. Continue working on intake lift arm

Day 17 - Cat lift arm motor

Goal: Complete attaching lift arm movement

Action Plan

1. Work on the lift arm motor attachment

Accomplishments

- Finished putting arm motor

Challenges

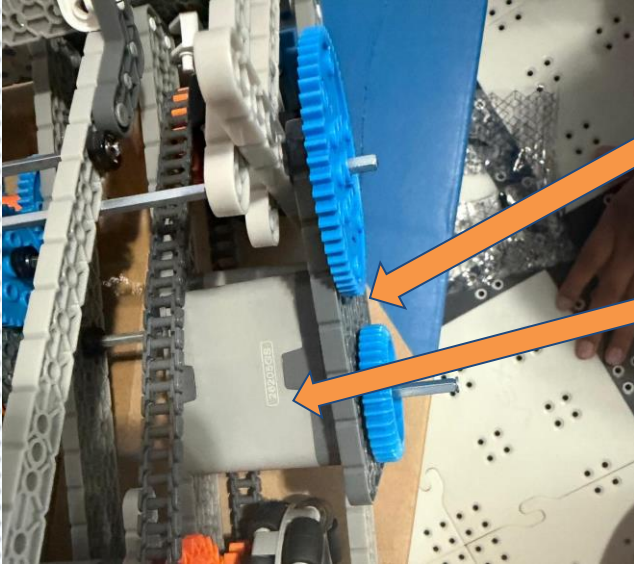
- Finding the correct placement of gears and motors for the lift

Lessons learned

Next steps

1. Continue working on lift arm

Day 17 - Challenges and Solutions



Finding the Right Place for the Arm Motor

Problem: Finding the right place for the arm motor and gears was challenging. If we put the motor too far back, it put too much strain on the chain. If we moved it too far forward, it touched the drivetrain sprocket. Since we did not have much space, we tried different gear combinations.

Solution: After trying different ideas, multiple combinations of different gears so they all align we found a solution when we used three gears: a **12-tooth gear connected to a 36-tooth gear, which connected to a 48-tooth gear**. With motor on 12 tooth gear we could put motor as right place.



Gears Skipping

Problem: After we put the gears, sometimes the arm would move and the gears would skip. The 36-tooth and 48-tooth gears were close to the edge of their teeth, so they could wobble and move out of place.

Solution: We added a plate to support the gears and keep them from wobbling or moving around. This helped the gears stay lined up and work better.



Day 18 - Driving Test

Goal: Complete the bot and start driving tests

Action Plan

1. Team 1: Working on the rubber band to fire the catapult
2. Team 2: Work on autonomous coding

Accomplishments

- We attached the rubber bands to the catapult.
- We started testing the robot.
- We tried different rubber bands to see which worked best.
- We replaced the rubber band shaft to be longer.

Challenges

- Wiring the bot, some places were hard to reach
- Catapult did not always shoot where we wanted.
- We tried many rubber bands to reach Level 2 and Level 3.
- It was hard to find the best place for the robot to shoot from.
- The rubber bands kept slipping off because the shaft was too short.

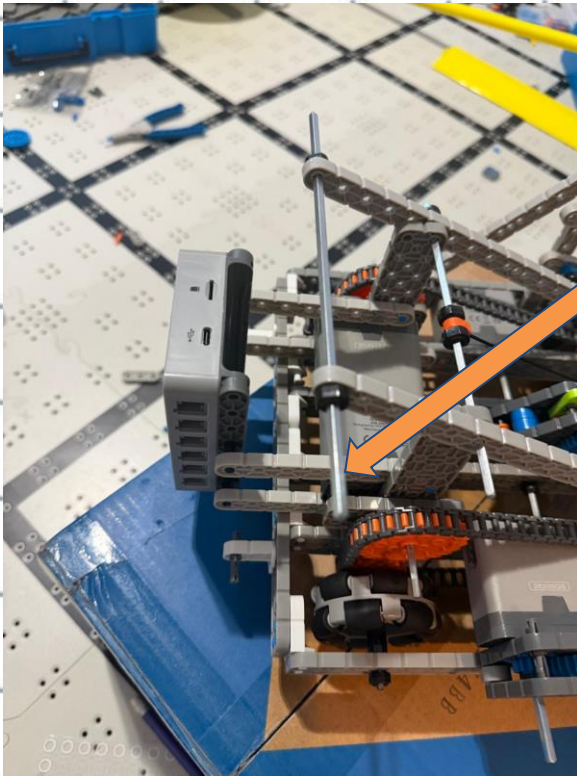
Lessons learned

- We learned that we have to test many times before finding what works.

Next steps

1. Keep testing different rubber band setups
2. Find the best shooting spots for levels 2 and 3.
3. Practice shooting so we can score more points.
4. Work on autonomous code

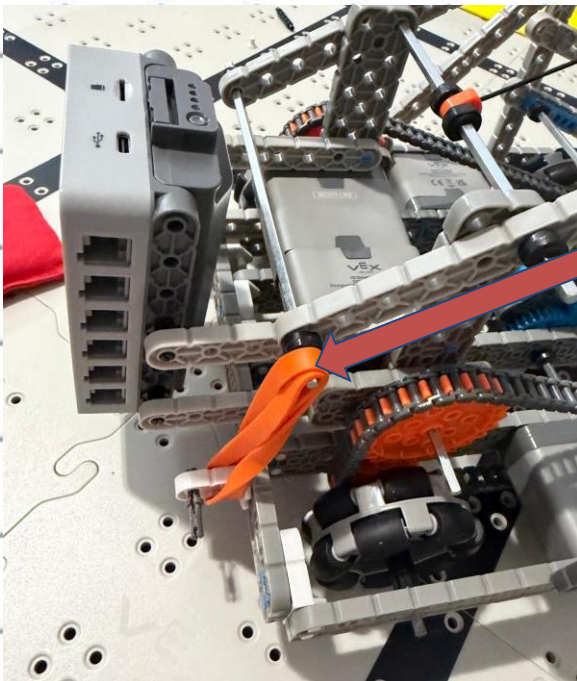
Day 18 - Challenges and Solution



Rubber band shaft

Problem: The rubber bands kept slipping off because the shaft was too short.

Solution: We replaced it with a longer shaft so the rubber bands would stay on better and not slip off.



Rubber band tension

Problem: We wanted the catapult to reach Level 3, but some rubber band combinations were too weak and some were too strong.

Solution: We tested different numbers and positions of rubber bands until we found a combination that could shoot the bean bag to Level 3..

A lot of times bag would still bounce off from level 3, so we will continue to iterate.

Day 19 - Tweaks to score

Goal: Complete drivetrain and Catapult

Action Plan

Team 1: Work on tweaks to score better
Team 2: Work on autonomous coding

Accomplishments

- We added a stopper so the catapult would not go too far back.
- We designed two antennas to help us score better.
- We added supports to keep our robot slightly tilted.
- We improved our driving so the motor would not hit the field border.

Challenges

- Sometimes the catapult went too far back and got stuck.
- We tried different designs to improve our scoring.
- The motor sometimes hit the field border.
- The standoff stopped us from reaching bean bags that went too far back.

Lessons learned

- We learned that small changes to our design can help us score better.
- We learned that fixing one problem can sometimes create another problem.

Next steps

1. Practice driving
2. Work on auton code

Day 20 - Cat Driving and Fixing

Goal: Keep driving and fixing problems we find with Cat

Action Plan

1. Practice driving to get better and find problems

Accomplishments

- Everyone got a turn to practice driving
- We solved our problems and made our robot better

Challenges

- Motor would hit the border of the field
- We wanted to improve our scoring and keep our robot slightly tilted.

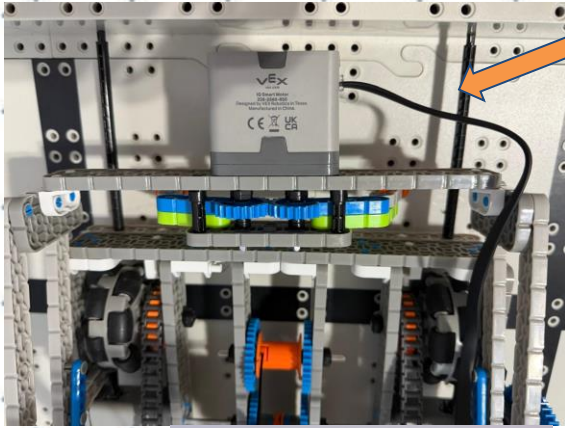
Lessons learned

- Sometimes finding problems helps you find solutions

Next steps

1. Continue to practice driving
2. Continue making small changes when we find problems

Day 20 - Problems and Solutions



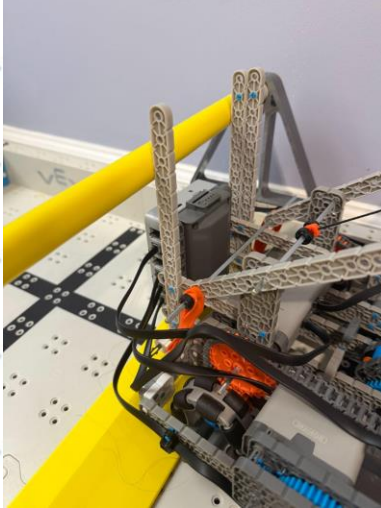
Motor hitting field boundary

Problem: Motor would hit the border of the field

Solution: We added stand off to prevent that happening

Problem: Sometime bean bag will go all the way back and we will not be able to reach the bean bag

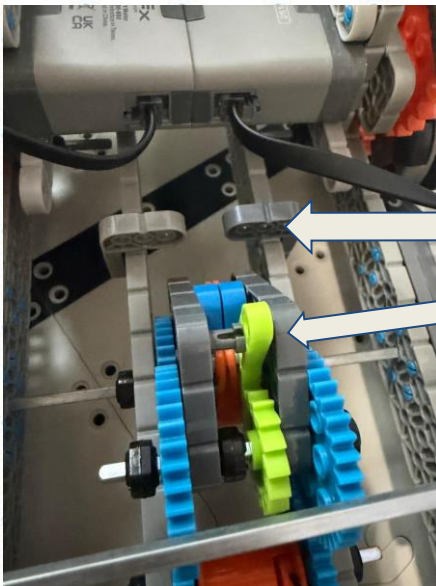
Solution: Remove stand off. For the motor hitting field, we learned to drive better and keep arm up.



Improve Scoring

Problem: We wanted to improve our scoring and keep our robot slightly tilted.

Solution: We tried different designs and came up with two antennas to help us score better. We also added two supports on one side and one on the other to keep our robot slightly tilted.



Catapult Getting stuck

Problem: Sometimes the catapult mechanism went too far back and got stuck. It would not come forward again.

Solution: We added a stopper to prevent the catapult mechanism from going too far back and getting stuck.

Day 21 - Fix cat arm

Goal: Fix cat arm

Action Plan

Team 1: Practice driving and loading
Team 2: Work on autonomous coding

Accomplishments

- Finally fixed cat arm after 4 tries!

Challenges

- Gears would not align
- Gears would skip

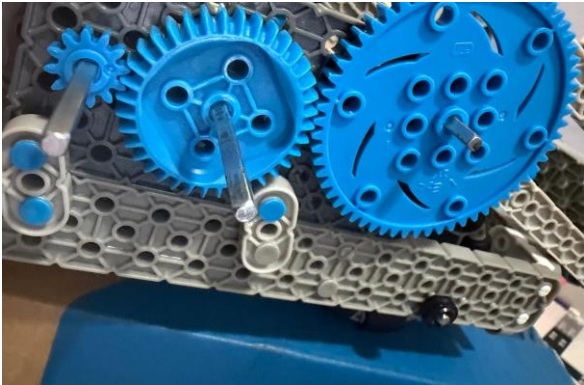
Lessons learned

- Sometimes you have to try a lot of times
- Everyone thinking about a problem can get a lot more ideas than just one person doing it

Next steps

1. Keep working on driving and autonomous coding

Day 21 - Challenges and Solutions

Solution Options	Pros / Cons / Notes
	Iteration #1 UNSUCCESSFUL We had to adjust the arm so bean bag would go in the catapult properly, but when we did that we had to redo the gears. Tried many different gear combinations for the arm gear. Motor set perfectly, but gears will not align.
	Iteration #2 UNSUCCESSFUL Added one more gear, it worked well, but during driving it will sometime skip, so we had to find another ways to do it.
	Iteration #4 SUCCESS Brainstormed as a team. Moved around the arm base, and then used 12:36:64 gear ration and it worked.

Day 22 - Autonomous Driving Test

Goal: Mostly finalize autonomous code for competition

Action Plan

1. Run the robot to see what problems the autonomous code has, then fix those problems

Accomplishments

- We added a bumper to stop the catapult
- We kept working on our drivetrain problems through many different solutions

Challenges

- Catapult would not stop at the right place
- One side of the drivetrain stopped working

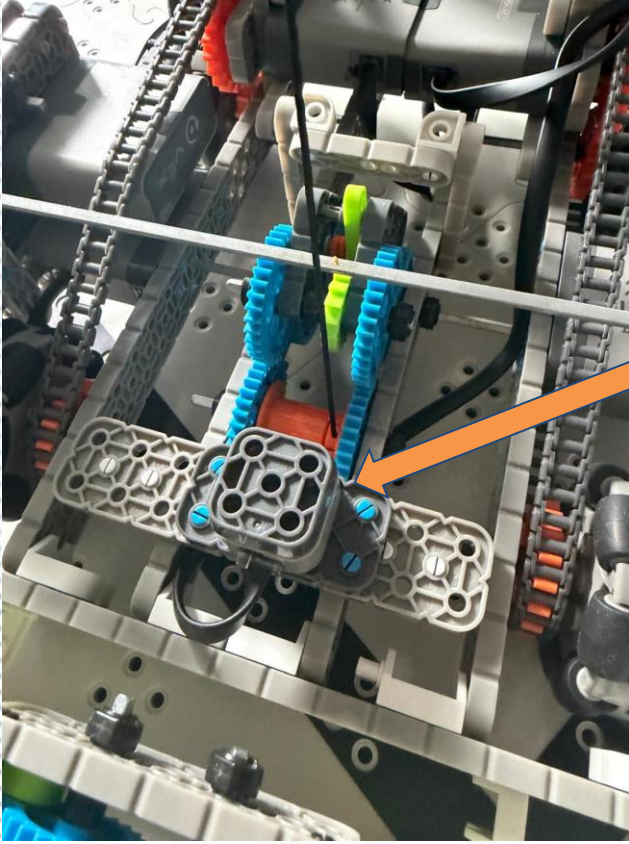
Lessons learned

- A lot of coding is trying, seeing what doesn't work, then trying again until it works how we want it to

Next steps

1. Practice driving for competition

Day 22 - Challenges and Solutions



Catapult Stopping

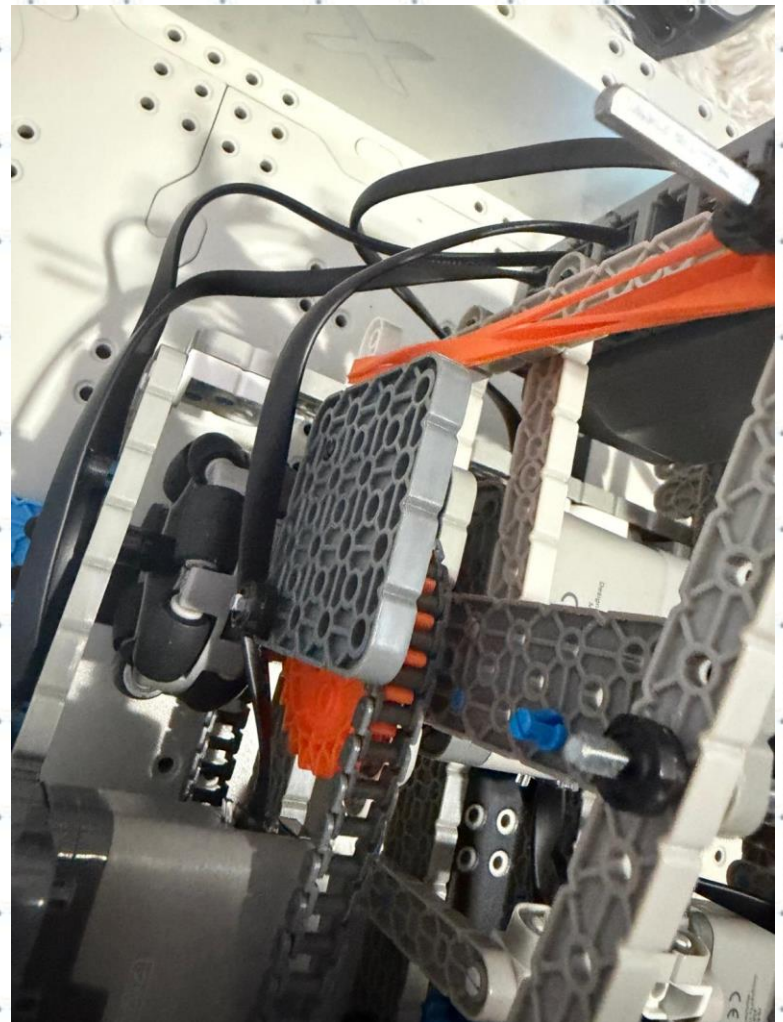
Problem: During autonomous driving we couldn't figure out how to stop catapult. We tried different turns and degree turns. It worked, but did not work consistently.

Solution: In the .Flop we had used the bumper and team came up with the idea to put the bumper, so when catapult hits the bumper it will stop the catapult.

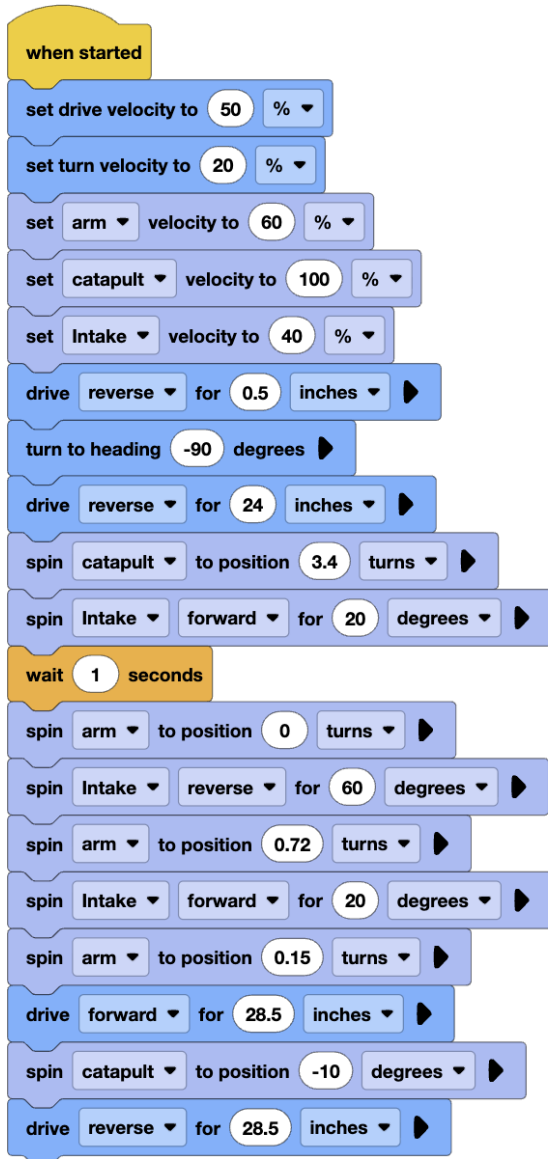
One side of drivetrain stopped working

Problem: During driving test one of the wheels would stop driving. We checked motor, cables, and everything was good. The shaft came out slightly. We reinserted shaft and it would do it again after driving for some more time.

Solution: We put a bar and attached to the side of brain support, so it will not move the drive shaft. This was not an easy solution, we had to try many different solutions.

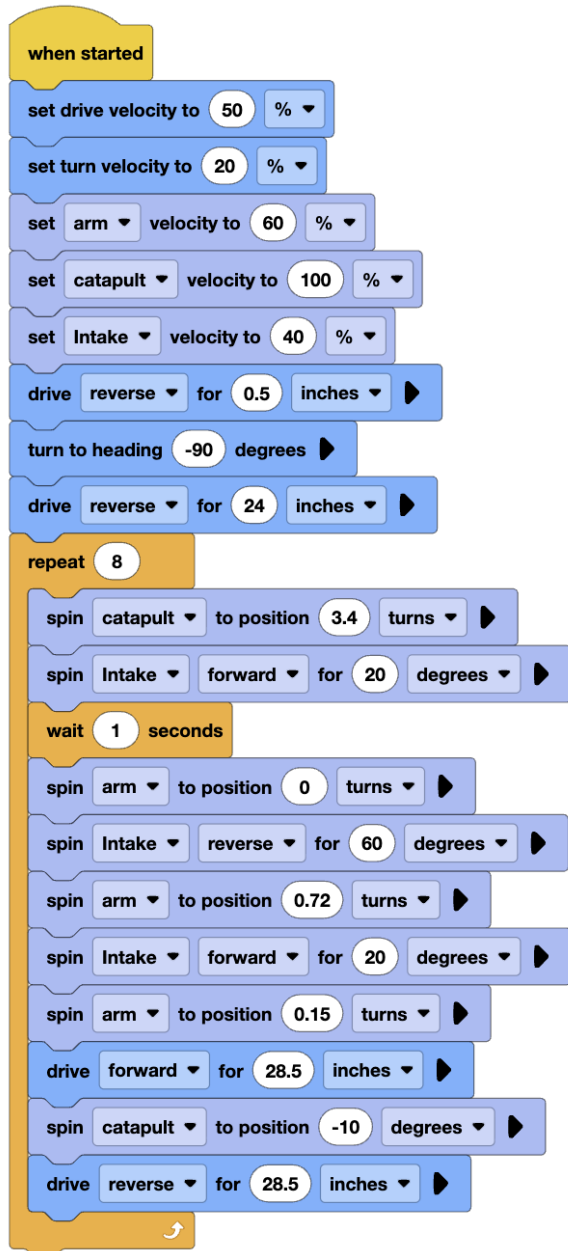


Autonomous Code



After many many trials, iterations, and trying different values, we made a simple code that would throw one bean bag.

Autonomous Code Contd...



After many many trial, iterations and trying different values, we made a simple code that would do back and forth 8 times and put 8 bean bags. We did not time this, but we still have to combine these codes, and then time it.

Autonomous Code Contd...

Combined code that does first loop since it has 90 degree turn, and then other 8 loops back and forth without turns.

We added code for the bumper press.

```
when started
  set drive velocity to 50 %
  set turn velocity to 20 %
  set arm velocity to 60 %
  set catapult velocity to 100 %
  set Intake velocity to 40 %
  drive reverse for 0.5 inches
  turn to heading -90 degrees
  drive reverse for 24 inches
  spin catapult to position 3.4 turns
  spin Intake forward for 20 degrees
  wait 1 seconds
  spin arm to position 0 turns
  spin Intake reverse for 60 degrees
  spin arm to position 0.72 turns
  spin Intake forward for 20 degrees
  spin arm to position 0.15 turns
  drive forward for 28.5 inches
  spin catapult to position -10 degrees
  drive reverse for 28.5 inches

  wait until bumper12 pressed?
  stop catapult
  set Intake stopping to hold
  spin Intake forward for 20 degrees
  wait 1 seconds
```

```
spin catapult to position -10 degrees
drive reverse for 28.5 inches

repeat 8
  spin catapult to position 3.4 turns
  spin Intake forward for 20 degrees
  wait 1 seconds
  spin arm to position 0 turns
  spin Intake reverse for 60 degrees
  spin arm to position 0.72 turns
  spin Intake forward for 20 degrees
  spin arm to position 0.15 turns
  drive forward for 28.5 inches
  spin catapult to position -10 degrees
  drive reverse for 28.5 inches
```

Days 23 - 24 - Competition Prep

Goal: Practice driving, loading, and coding

Action Plan

1. Take turns practicing driving, loading, and working with autonomous code

Accomplishments

- Best scorer scored twice as many points as last practice
- Everyone is improving with practice

Challenges

- We are all new drivers so we don't always do very well

Lessons learned

- We can keep improving every time

Next steps

1. Practicing every evening this week for competition on Saturday!